

Adroit Technologies

Protocol Driver

Name	DSC EnvisaLink Ethernet Driver
DLL	DSCEnv



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1. Overview

This document describes the setup of, and the communication protocol implemented in the Adroit protocol driver, with the following logic unit: DSC EnvisaLink.

2. Revisions

2.1. Document Revision

Rev.	Date	Author	Comment	Approval
1.0	14-Dec-2022	J. Duncan	New Format	D. Jordaan
1.1	17-Jan-2023	K. Robinson	Changes to format, updated image resources. Added Appendix B DCOM Hardening. Removed Windows XP section	D. Jordaan
2.0				
3.0				
4.0				
5.0				
6.0				

2.2. DLL Revision

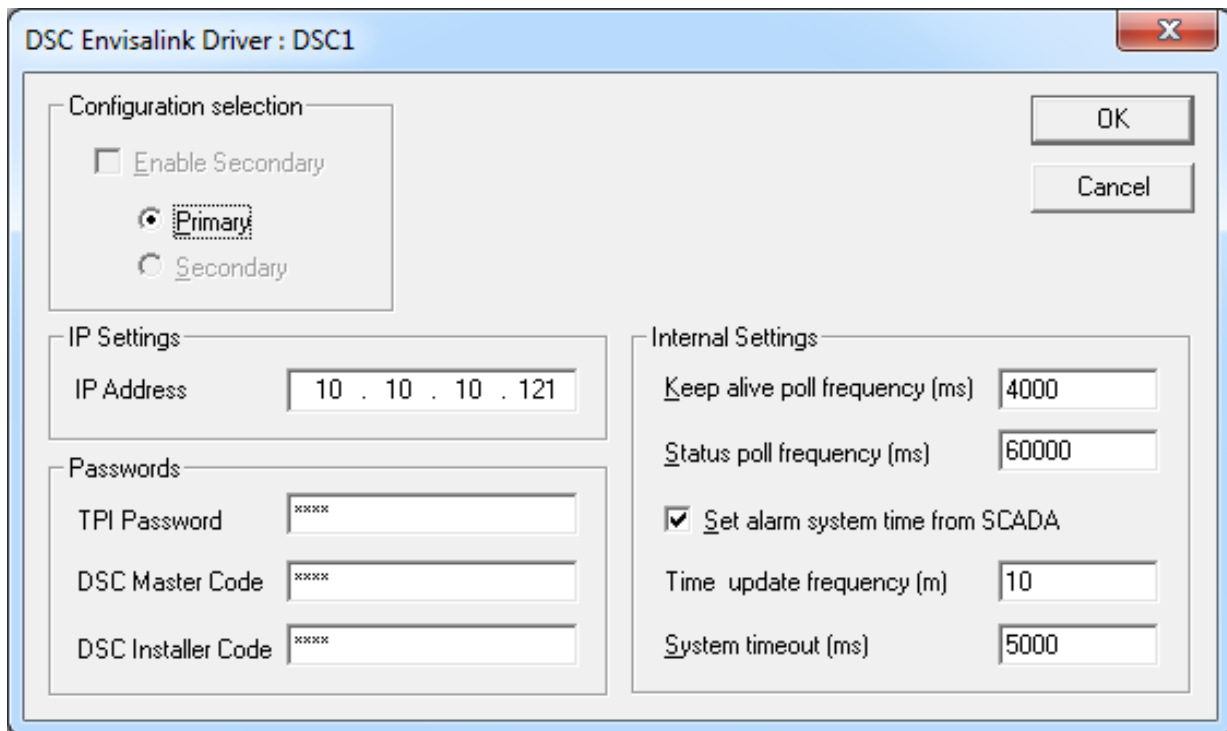
Rev.	Date	Changes
	10-Feb-2012	First Release notes!

3. Wiring

Standard Ethernet connection

4. Device Configuration

4.1. Device Configuration



4.1.1. Configuration Selection (not available)

Enable Secondary – [Not implemented] Enables secondary configuration for dual redundant Controllers. This setting is disabled for this driver.

Primary / Secondary – [Not implemented] Select the primary or secondary button depending on which connection you wish to configure.

This setting is disabled for this driver.

4.1.2. IP Settings

IP Address – The IP Address that will be used for communicating with the controller.

4.1.3. Passwords

TPI Password – Enter your EnvisaLink webpage password here

DSC Master Code – Enter the DSC alarm panel's master code. This is used to arm and disarm the alarm panel.

DSC Installer Code – Enter the DSC alarm panel's installer code. This is used when configuring the alarm panel.



PLEASE NOTE

Passwords are encrypted using 16-bit key encryption before being stored in the registry.

4.1.4. Internal Settings

Keep alive poll frequency (ms) – This is the time delay between keep alive polls. This value is measured in milliseconds and must be between 1000 and 10000.

Status poll frequency (ms) – This is the time delay between status polls. This value is measured in milliseconds and has a minimum of 10000ms.

Set alarm system time from SCADA – If this checkbox is checked, the alarms panel's system time will be set with the time from the Adroit agent server.

Time update frequency (m) – The time on the alarm panel may drift, if "Set alarm system time from SCADA" is check, the alarm panel's system time will be updated with this frequency.

Timeout (ms) – If the driver does not receive a valid response from the device within this period (measured in milliseconds), the port will be closed. Note: this value must be greater that the "keep alive poll period" in the "Internal Settings" section.



PLEASE NOTE

This driver has an internal thread which handles the scanning. Changing these settings changes the frequency in which the scanning thread adds the Adroit jobs to the read queue.

5. Supported Addresses

The DSC EnvisaLink Adroit Ethernet driver supports the following data address types:

Address	Description	Min Addr	Max Addr	Addressing example	Supported Adroit Data types	Read/Write
PAR	Partition status request	1	8	PAR 1 PAR 8	Integer	Read
ZON	Zone status request	1	64	ZON 1 ZON 64	Integer	Read
ALM	Alarm status request	1	5	ALM 1 ALM 5	Digital Integer	Read
GEN	General status request	1	4	GEN 1 GEN 4	Digital Integer	Read
TBL	Trouble LED request	1	8	TBL 1 TBL 8	Digital Integer	Read
PAD	Partition Arm/Disarm	1	8	PAD 1 PAD 8	Integer	Write
KEY	Keypad Sequence	1	8	KEY 1 KEY 8	String	Write
PAA	Panic Alarm Activate	0	0	PAA	Integer	Write
PGM	Programmable output activate	1	4	PGM1 PGM4	Integer	Write



PLEASE NOTE

Please read the Address Descriptions section before using these addresses.

5.1. Address Descriptions

The addresses are either read only or write only. Read only addresses are sent from the driver to Adroit in an unsolicited manner. Write only addresses will be cleared (for example, to 0 if digital or integer type and to "" if it is a string type) once the write is complete.

5.1.1. Partition status request (PAR)

[Read only address] The value sent to Adroit when scanning this address represents the state of the partition being scanned.

Value	State
1	Partition ready
2	Partition not ready
4	Partition armed in away mode
8	Partition armed in stay mode
16	Partition armed in zero entry away mode
32	Partition armed in zero entry stay mode
64	Partition in alarm
128	Partition in exit delay
256	Partition in entry delay

5.1.2. Zone Status Request (ZON)

[Read only address] The value sent to Adroit when scanning this address represents the state of the zone being scanned.

Value	State
1	Zone restored
2	Zone Open
4	Zone Fault
8	Zone Tamper
16	Zone Alarm

5.1.3. Alarm Status Request (ALM)

Each address references a different type of alarm. The following table indicates what type of alarm each addresses refers to.

Address	Alarm Type
ALM 1	Duress alarm state
ALM 2	Fire alarm state
ALM 3	Auxiliary key alarm state
ALM 4	Panic alarm state
ALM 5	Smoke alarm state

[Read only address] The value sent to Adroit when scanning this address represents the state of the alarm being scanned.

Value	State
0	Not alarmed
1	In Alarm

5.1.4. General status request (GEN)

[**Read only address**]. Each address references a unique general status. The table below indicates what each address references to.

Address	Status Type
GEN 1	Panel battery trouble, battery low or battery has been removed.
GEN 2	Panel AC trouble, AC power to the panel has been removed.
GEN 3	System bell trouble, an open circuit has been detected across the bell terminals.
GEN 4	General system tamper.

The value sent to Adroit when scanning these addresses represents the general state of the status type being scanned.

Value	State
0	Good status
1	Bad status

5.1.5. Trouble LED Request (TBL)

[**Read only address**]. The table below indicates the state of the trouble LED on the keypad of the partition being scanned (1-8).

Value	State
0	LED off
1	LED on

5.1.6. Partition Arm/Disarm (PAD)

[**Write only address**]. The table below describes the resultant action on the partition scanned (1-8) when writing the following values from Adroit:

Write Value	Result of write
1	Arm partition in away mode
2	Arm partition in stay mode
3	Arm partition with zero entry delay
4	Disarm partition with user code entered in the dialog

5.1.7. Keypad Sequence (KEY)

[**Write only address**]. This address allows the driver to simulate a keystroke sequence as if it were being pressed on a specific partition's (1-8) keypad.

Only the following characters can be sent:

'1 2 3 4 5 6 7 8 9 * #'

If any other chars are found in the character sequence, the write will fail.

5.1.8. Panic Alarm Activate (PAA)

This is a write only address. The following table indicates which panic alarms will be triggered when certain values are written to this address.

Write Value	Result of write
1	Trigger fire alarm
2	Trigger ambulance alarm
3	Trigger police alarm

5.1.9. Programmable Output Activate (PGM)

[**Write only address**]. The address index (1-4) represents the PGM that will be output when written. Valid write values are 1-8. The write value is partitions which activates the programmable output.

6. Protocol Definition and Considerations

6.1. DSC EnvisaLink Message Structure

Command Code	Data	Checksum	Carriage return & line feed
CCC	DDD...DDD	KK	\r\n

Command Code:

This three digit code when sent to the device issues a command, i.e. tells the device what to do. However when receiving a message from the device, this code tells the driver what type of information is being received.

Data:

This part of the message is variable length depending on the command code sent, and contains the data required for the command code.

Checksum:

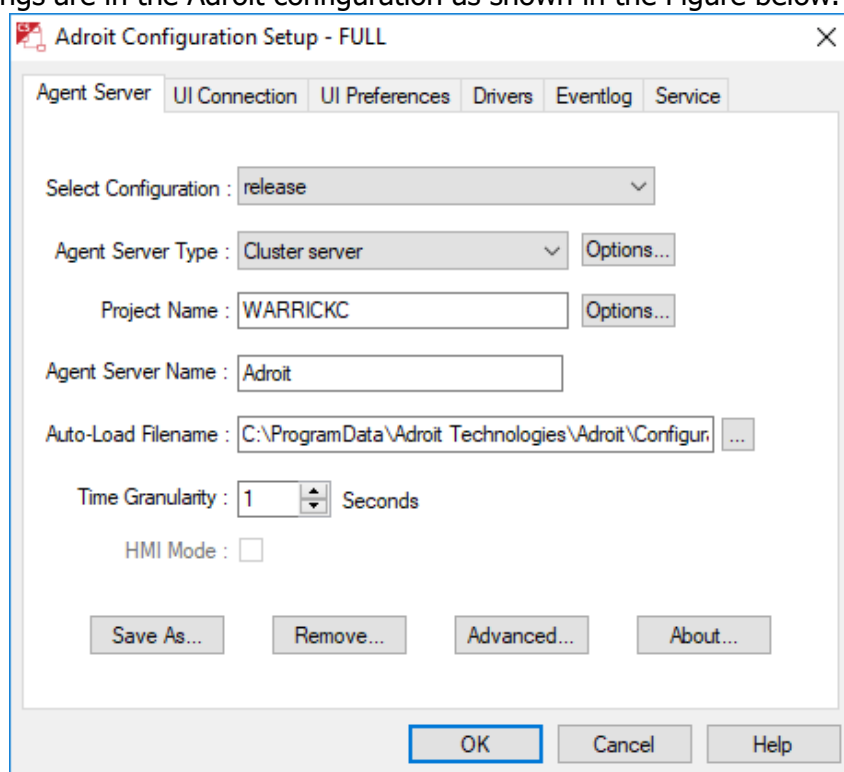
The checksum is two characters wide. The checksum is calculated by adding the hexadecimal value of all command and data digits, then truncating the result to 8 bits.

Carriage return & line feed:

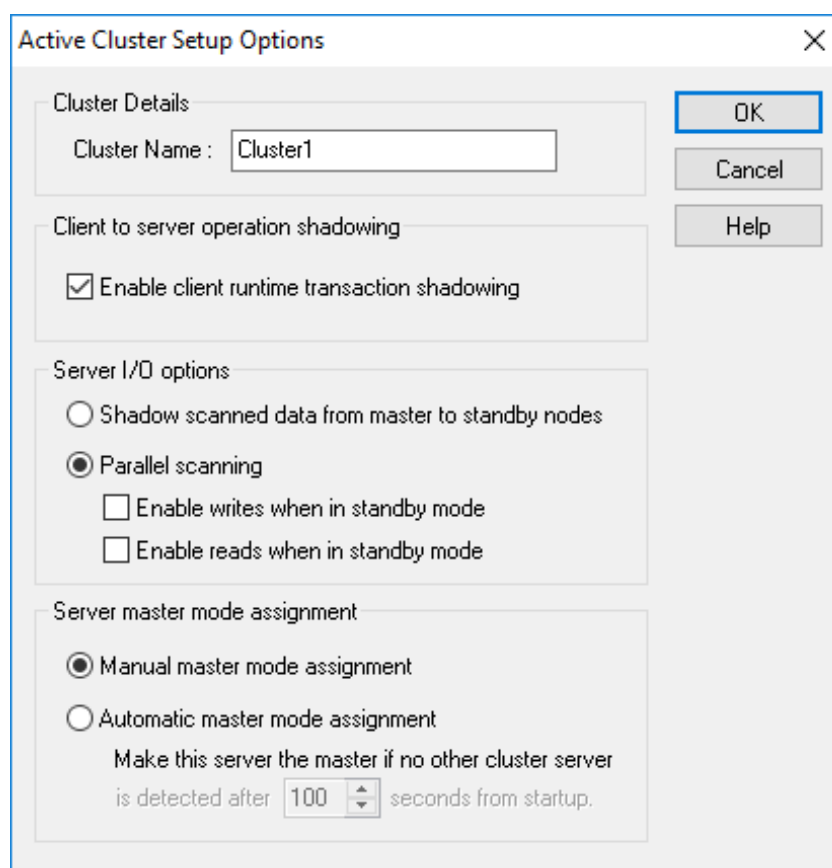
The carriage return and line feed control characters are appended to the end of each message.

6.2. Cluster Awareness Settings

Cluster settings are in the Adroit configuration as shown in the Figure below.



To ensure the standby port is closed when the Master is communicating, open the Adroit Configuration Setup. The standby machine must be in shadow mode or in parallel mode (with both enable writes and enable read checkboxes unchecked) as shown below:



7. General Notes and Troubleshooting

7.1. Contact Details

Email: support@adroit.co.za and drivers@adroit.co.za

Phone: +27 11 658-8100

Web: www.adroit.co.za

7.2. Adroit Technologies Knowledge Base

For additional information, please refer to our Knowledgebase website:

[Latest Drivers topics - Features, discussions, tips, tricks, questions, problems and feedback \(adroittechnologiesautomation.com\)](http://adroittechnologiesautomation.com)